SCSE 2123 SOFTWARE REQUIREMENTS ENGINEERING

Topic 2: FUNDAMENTAL PRINCIPLES OF REQUIREMENTS ENGINEERING

Innovating Solutions

1

Contents

2.1 Overview of Principles
2.2 The Principles Explained
2.3 Stakeholders (Principle 2)
2.4 System and Context (Principle 4)
2.5 Sustainability Analysis for System Domains - Environmental, social, and economic contexts

www.utm.my

2

Objectives

- Students will be able to **explain** the fundamental principles of Requirements Engineering
- Students will be able to **apply** the principles of Requirements Engineering to identify and relate stakeholders, system boundaries, and context elements in a given application domain
- Students will be able to **analyze** how sustainability aspects (environmental, social, and economic) influence the identification and prioritization of system requirements within a defined context

www.utm.my

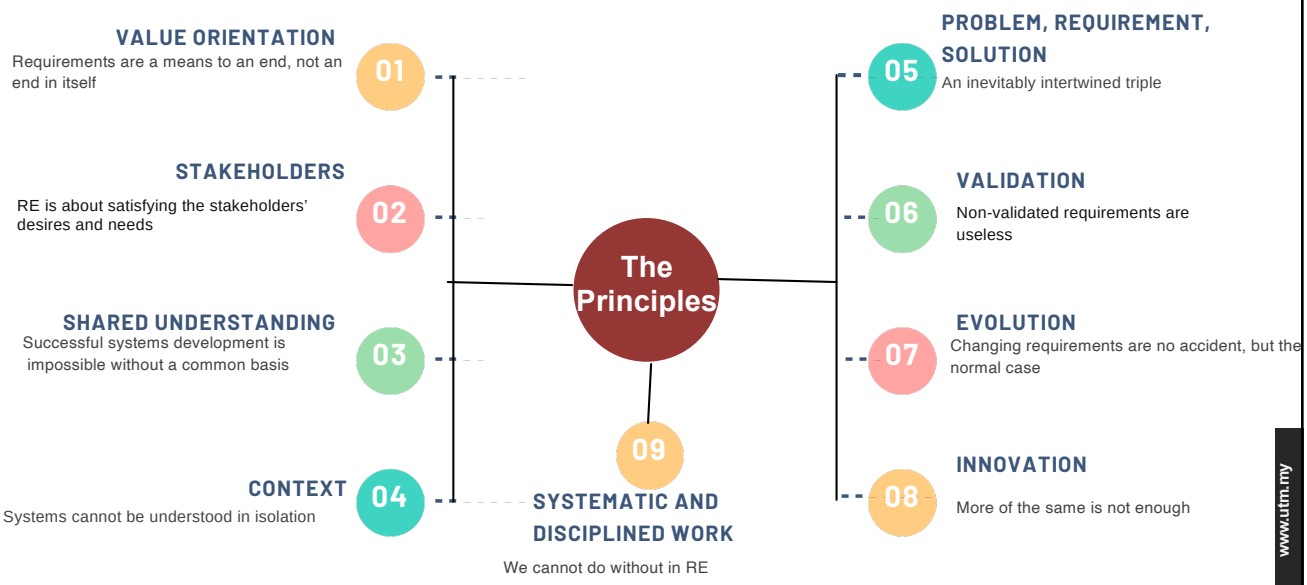
3

2.1 OVERVIEW OF THE PRINCIPLES

www.utm.my

4

Requirements Engineering Principles



www.utm.my

5

2.2 THE PRINCIPLES EXPLAINED: PRINCIPLE 2 & 4

www.utm.my

Innovating Solutions

6

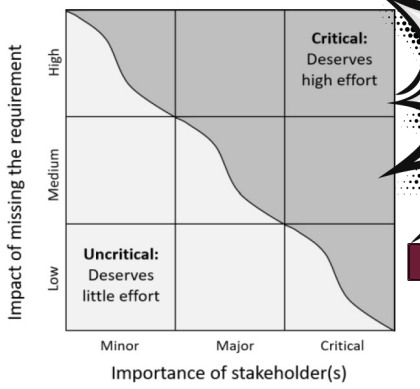
2. Stakeholders

- RE is about understanding the stakeholders' desires and needs
- Properly dealing with stakeholders is a core task of RE
- Every stakeholder has a role in the context of the system to be built e.g., user, client, customer, operator, or regulator
- Depending on the RE process used, developers and testers of a system can also be stakeholders

2. Stakeholders

- Stakeholders may have different needs and viewpoints, which may result in conflicting requirements.
- Identify inconsistencies and conflicts between requirements of different stakeholders and resolve these.

2. Stakeholders



Stakeholder classification based on inflluence of stakeholder

Critical

Not considering these stakeholders will result in severe problems and probably make the system fail or render it useless

Major

Not considering these stakeholders will have an adverse impact on the success of the system but not make it fail

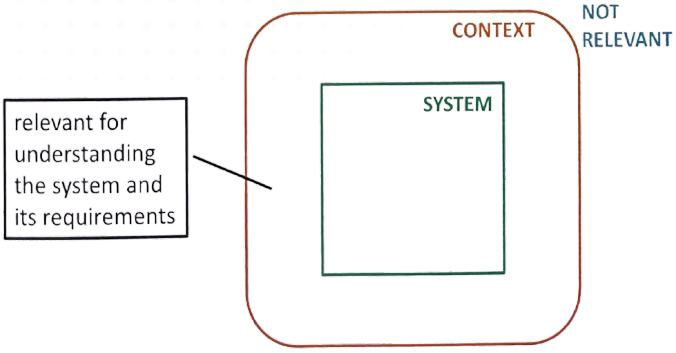
Minor

Not considering these stakeholders will have no or minor influence on the success of the system

Involving the right people in the relevant stakeholder roles is crucial for successful RE

4. Context

- Systems are embedded in a context.
- Without understanding that context, it is impossible to specify a system properly.



4. Context: Important Terms

- System context
- System boundary
- Context boundary
- Scope
- System context objects
- Interfaces
- Grey zone

www.utm.my

11

4. Context

Context **(definition)**

- In general: The network of thoughts and meanings needed for understanding phenomena or utterances.
- Especially in RE: The part of a system's environment being relevant for understanding the system and its requirements.

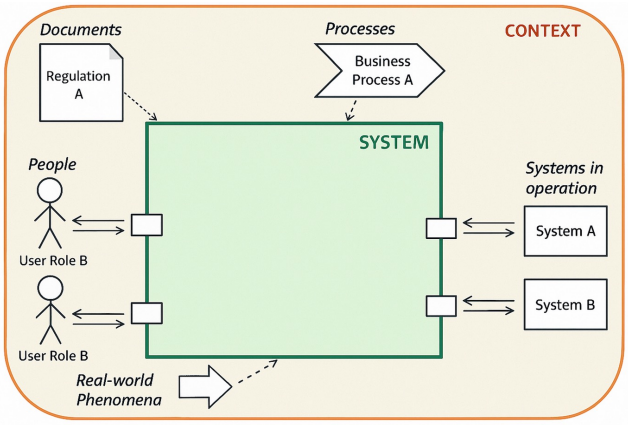
System Context **(definition)**

- The part of a system's environment that is relevant for the definition as well as the understanding of the requirements of a system to be developed.

www.utm.my

12

System and Context



NOT RELEVANT

System X

www.utm.my

13

System and Context

Context of a System

Accounting System (1)

E

Example: Develop an accounting system for the manufacturing industry in Germany.

In this context we have to consider

- relevant German laws and
- business needs of manufacturing industry.





All icons from [1]

© University of Duisburg-Essen, Prof. Dr. K. Pohl, Germany
For Academic Use Only.

www.utm.my

14

System and Context

Context of a System

Accounting System (2)

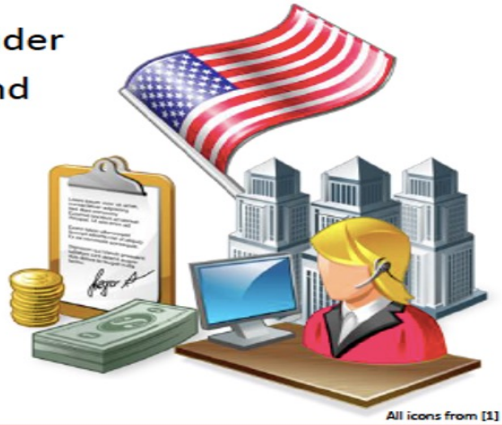
E

Example: Develop an accounting system for financial companies in the USA.

In this context we have to consider

- relevant US American laws and
- business needs of financial companies.





All icons from [1]

PALUNO The Ruhr Institute for Software Technology
© University of Duisburg-Essen, Prof. Dr. K. Pohl, Germany
For Academic Use Only.

www.utm.my

15

System and Context

Context of a System

Accounting System (3)

E

Example: Develop an accounting system for a shipping company in China.

In this context we have to consider

- relevant Chinese laws and
- business needs of shipping companies.





All icons from [1]

PALUNO The Ruhr Institute for Software Technology
© University of Duisburg-Essen, Prof. Dr. K. Pohl, Germany
For Academic Use Only.

www.utm.my

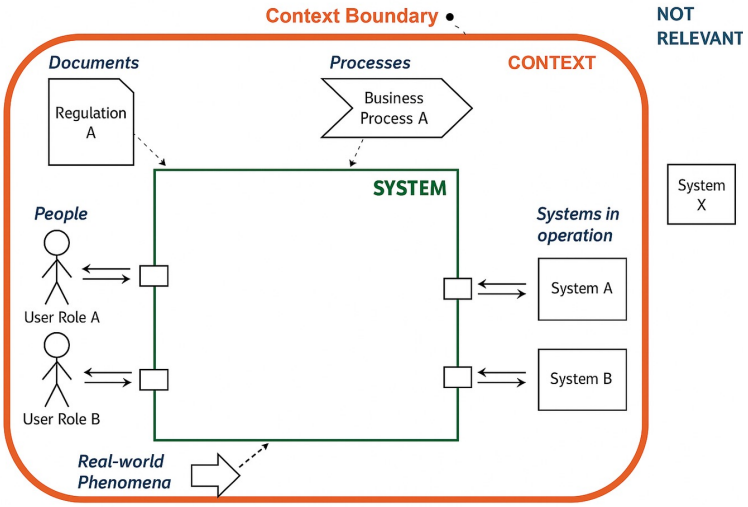
16

Context Boundary

Context Boundary (definition)

The boundary between the context of a system and those parts of the application domain that are irrelevant for the system and its requirements.

- The context boundary defines the scope of the environment that affects or interacts with the system.
- It includes people, external systems, processes, documents, and real-world phenomena that the system communicates with — but are not controlled by the system.
- Everything outside the context boundary is not relevant to the system or its requirements



www.utm.my

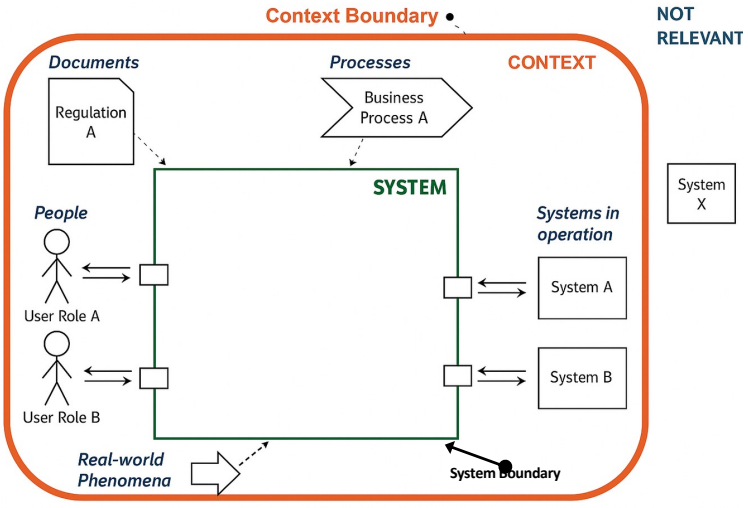
17

System Boundary

System Boundary (definition)

The boundary between a system and a surrounding context

- The system boundary delimits the system as it shall be after implementation and deployment.
- At the system boundary, the external interfaces between system and its context have to be defined.



www.utm.my

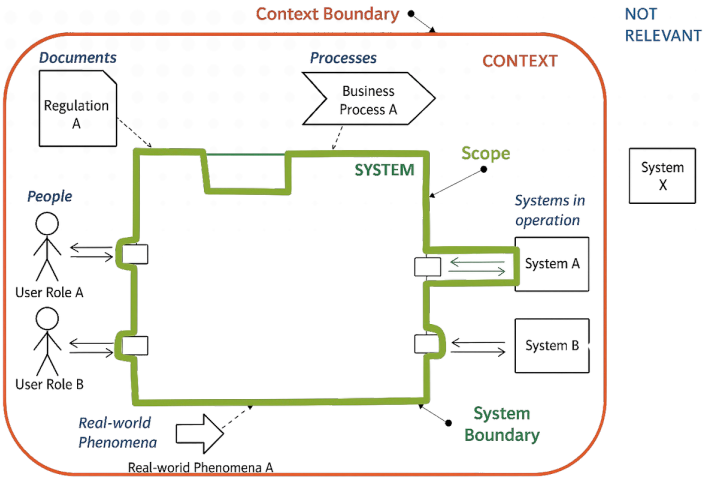
18

Scope

Scope (definition)

Aspects that can be changed and designed during system development

- Aspects belong to the environment, cannot be altered during development; constraints for the system to be developed



www.utm.my

19

System Context Objects

System Context Objects (definition)

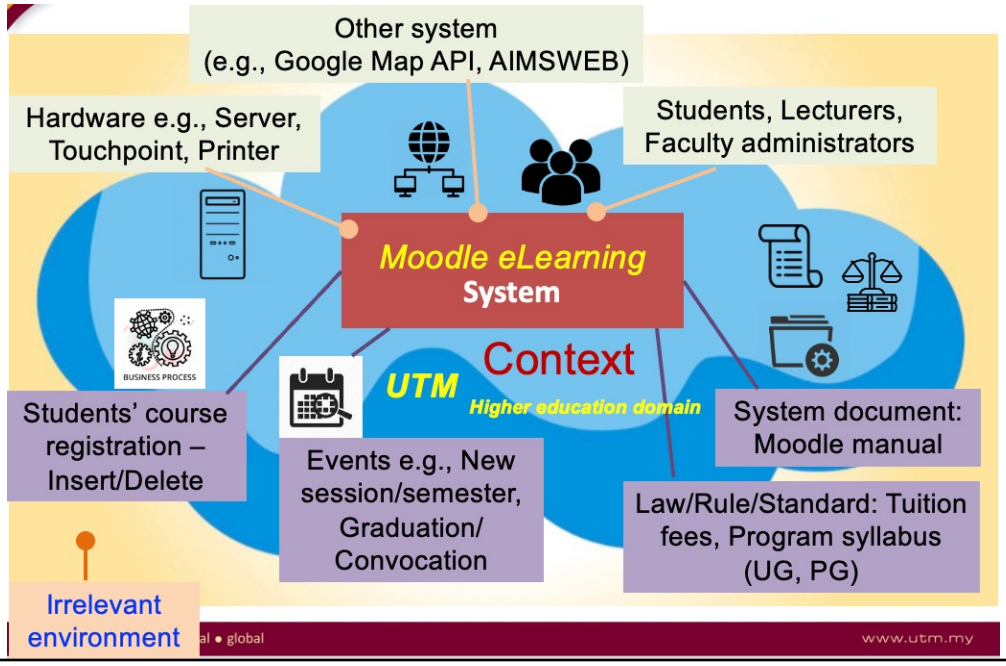
Relevant parts/aspects belong to the system environment that influence to the requirements

- People (stakeholders or groups of stakeholders)
- Systems in operation (other technical systems or hardware)
- Processes (technical or physical processes, business processes)
- Events (technical or physical)
- Documents (e.g., laws, standards, system documentation)

www.utm.my

20

System Context Objects Example

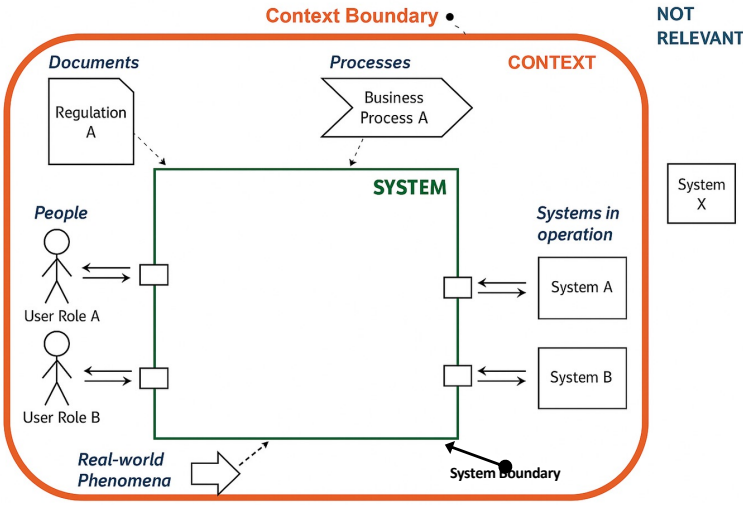


21

Interfaces

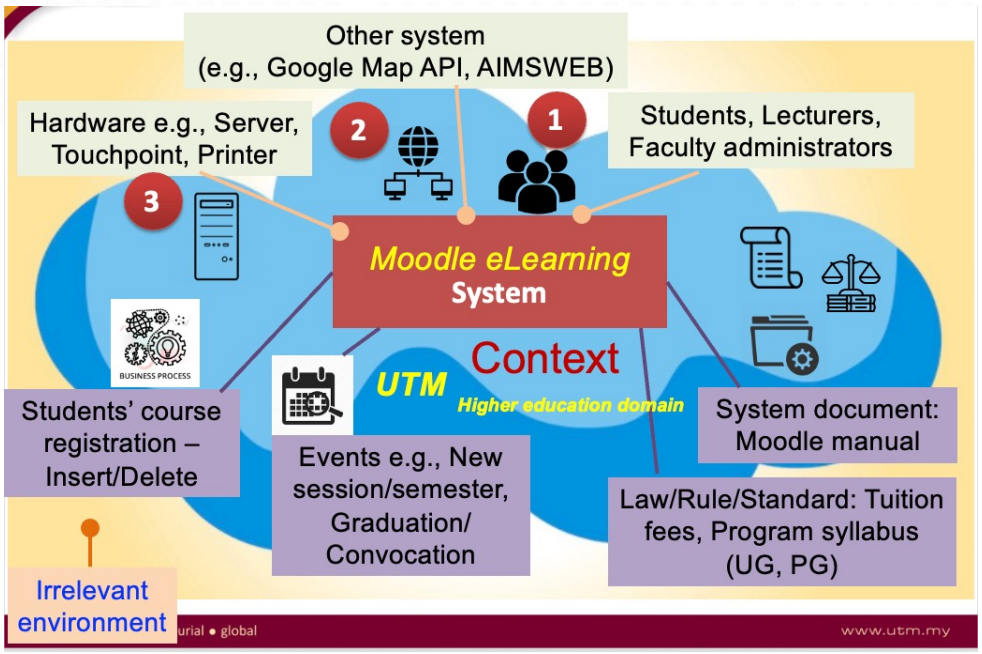
Interfaces (definition)

Interactions between the system and environment (i.e., human-people, software, hardware) that impose specific constraints@ additional sources of requirements



22

Interfaces



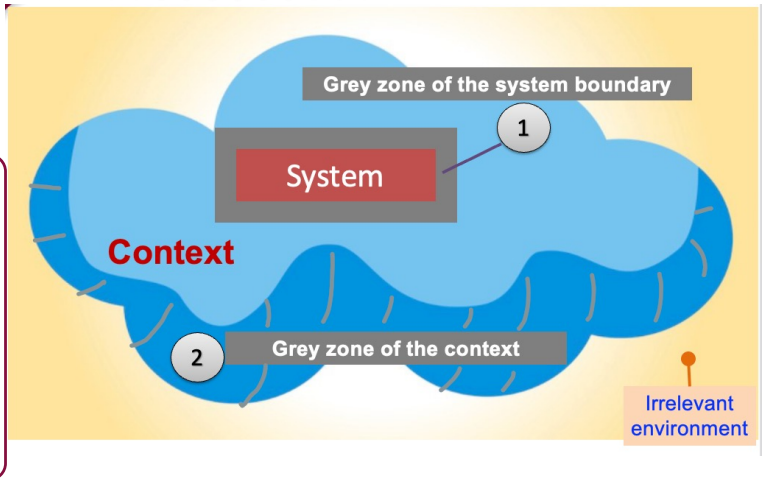
23

Grey Zones

Grey Zones (definition)

Identified aspects in the scope of environment (context) which is unclear/uncertain whether they have a relation to the system or not (vague separation)

- During requirements engineering, the system boundary and the grey zone boundary are typically unstable and change frequently.
  - System context objects might become part of the grey zone or the system,
  - Grey zone objects might become part of the system context or the system.
  - System objects might become part of the grey zone or the system context.



24

2.3 SUSTAINABILITY ANALYSIS FOR SYSTEM DOMAINS – ENVIRONMENTAL, SOCIAL, AND ECONOMIC CONTEXTS

www.utm.my

Innovating Solutions

25

Why Sustainability in System Context

- Systems operate within complex ecosystem that impacts sustainability factors.
- Systems influence environmental resources, social welfare, and economic outcomes
- Sustainability ensures long-term viability and ethical impact
- Integrating sustainability aligns systems with global development goals

www.utm.my

26

3 Pillars of Sustainability

Environmental Sustainability

- Focuses on minimizing environmental footprint.
- Example: Green data centers, low-energy architectures.

Social Sustainability

- Promotes inclusion, ethics, and community well-being.
- Example: Accessibility features, fair labor practices.

•Economic Sustainability:

- Ensures cost-effectiveness and financial longevity.
- Example: Maintenance optimization, lifecycle value.

Goal: When we design or improve any system (for example, a transport application, healthcare system, a warehouse robot, or campus lighting), we judge options against the three pillars and turn the findings into **requirements** and **acceptance criteria**.

www.utm.my

27

Examples of Environmental Sustainability Context: University Registration System

Focus: Reducing environmental impact through digital transformation and resource optimization.

Aspect	Example of system context	Justification
Paperless processes	Student Portal, Registration Module, and Document Management System exchange digital forms, e-signatures, and e-receipts.	Interaction between students, admin staff, and URS replaces physical paperwork
Energy-efficient operations	University Registration System Server hosted on a Green Cloud Service Provider with energy-efficient infrastructure.	The external cloud provider affects the system's environmental footprint
Reduced travel emissions	*Online Access Interface* allows students to perform all registration tasks remotely via the university's web platform.	*Students interact via the internet, reducing the need for on-campus visits.*
Digital notifications	Notification Subsystem sends course confirmations and updates through Email Gateway or Mobile App Push Service.	Replaces printed communication through connection to external messaging services

www.utm.my

28

Examples of Social Sustainability Context: University Registration System

Focus: Equity, inclusion, and accessibility for all users

Aspect	Example of system context	Justification
Accessibility and inclusion	User Interface Layer integrates Accessibility API for screen readers and multi-language support	Ensures equitable access for users with disabilities or language barriers.
User data privacy and trust	Authentication Service and Data Encryption Module ensure secure student information handling.	Interfaces with Identity Management System and external security standards
User support and engagement	*Helpdesk Chatbot and Feedback Module allow students to report issues or suggest improvements*	Enhances continuous interaction between users and system administrators
Transparency in communication	Dashboard Module displays real-time registration status and fee breakdowns	Promotes open interaction between the system and users, fostering institutional trust.

www.utm.my

29

Examples of Economic Sustainability Context: University Registration System

Focus: Optimizing cost efficiency, resource allocation, and long-term viability.

Aspect	Example of system context	Justification
Cost reduction	Automated Registration Workflow reduces administrative workload and manual processing costs	The system optimizes internal operations and resource allocation
Scalable infrastructure	Cloud Hosting Platform provides scalable pay-as-you-go resources	The system scales dynamically with student intake, reducing over-provisioning costs
Digital payment integration	*E-Payment Gateway enables online fee transactions through Bank APIs or E-Wallet Services.*	Links the system with external financial systems for seamless transactions
Long-term maintainability	Modular Architecture allows incremental updates and integration with new technologies	Sustains system evolution without major reinvestment

www.utm.my

30

THANK YOU

univteknologimalaysia utm.my utmofficial

31